

Independent Validation of Transgenerational Inheritance of Learned Pathogen Avoidance in *Caenorhabditis elegans*

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eLife Assessment

This **valuable** study concerns a model for transgenerational epigenetic inheritance, the learned avoidance by *C. elegans* of the PA14 pathogenic strain of *Pseudomonas aeruginosa*. A recent study questioned whether transgenerational inheritance in this paradigm lacks robustness. The authors of this study have worked independently of the group that reported the original phenomenon and also independently of the group that challenged the original report. With **solid** data, this study independently validates findings previously reported by the Murphy group, confirming that the paradigm is reproducible elsewhere. The reviewers also appreciated the information on reagent sources used by different groups. The present study is therefore of broad interest to anyone studying genetics, epigenetics, or learned behavior.

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Abstract

Transgenerational inheritance of learned behaviors remains a controversial topic in biology. The Murphy lab previously demonstrated that *C. elegans* exposed to pathogenic *Pseudomonas aeruginosa* (PA14) not only learn to avoid this bacterium but also transmit this avoidance behavior to untrained offspring. A recent study has challenged these findings, questioning the robustness of this phenomenon. Here, we independently validate the transgenerational inheritance of PA14 avoidance in *C. elegans*. Adapting the protocol described by the Murphy lab, we show that worms trained on PA14 develop significant avoidance that persists in F1 and F2 generations, though with decreased strength after P0. Our results provide independent confirmation of transgenerational inheritance of PA14 avoidance through the F2 generation, in agreement with previous findings from the Murphy lab and in contrast to a recent study by the Hunter group that failed to detect this phenomenon beyond the F1 generation.

Introduction

Caenorhabditis elegans inhabits soil environments where it encounters diverse bacteria, including both nutritional and pathogenic species (Schulenburg and Félix, 2017 [↗](#); Neher, 2010 [↗](#)). This nematode has evolved sophisticated sensory mechanisms to distinguish beneficial from harmful bacteria (Kim and Flavell, 2020 [↗](#); Guillermin, 2018 [↗](#)). Interestingly, naïve *C. elegans* initially prefer the pathogenic bacterium *Pseudomonas aeruginosa* (PA14) over the non-pathogenic laboratory strain *Escherichia coli* (OP50) but rapidly learn to avoid PA14 following exposure (Zhang and Bargmann, 2005 [↗](#)).

The Murphy lab first demonstrated that this learned avoidance behavior can be transmitted to progeny, persisting through the F2 generation (Moore et al. 2019 [↗](#); Kaletsky et al. 2020 [↗](#); Moore et al. 2021; Sengupta et al., 2024 [↗](#)). This transgenerational epigenetic inheritance (TEI) has been linked to small RNA pathways and the dsRNA transport proteins SID-1 and SID-2. Recently, Sengupta et al. (2024) [↗](#) further identified a specific small RNA responsible for inducing transgenerational inheritance of learned avoidance.

However, Gainey et al. (2025) [↗](#), representing the Hunter group, reported that while parental and F1 avoidance behaviors were evident, transgenerational inheritance was not reliably observed beyond the F1 generation under their experimental conditions. Critically, Kaletsky et al. (2025) [↗](#) demonstrated that omission of sodium azide during scoring can completely abolish detection of inherited avoidance, revealing that this key methodological difference may explain the conflicting results between laboratories. The use of sodium azide to immobilize worms at the moment of initial bacterial choice appears essential for capturing the inherited behavioral response. These findings highlight how seemingly minor methodological variations can dramatically impact detection of transgenerational inheritance and underscore the need for independent replication using standardized protocols. Here, we adapted the protocol established by the Murphy group, maintaining the critical use of sodium azide to paralyze worms at the time of choice, to test whether parental exposure to PA14 elicits consistent avoidance in subsequent generations. Our study specifically focuses on the transmission of learned avoidance through the F2 generation, beyond the intergenerational (F1) effect, because this is where divergence between published studies begins. Our goal was to evaluate whether the reported transgenerational signal persists across two generations without reinforcement.

Results and Discussion

We investigated whether *C. elegans* exposed to PA14 exhibit learned avoidance behavior that persists across generations. Naïve worms showed a preference for PA14 over OP50, with a negative choice index (approximately -0.4). After 24 hours of exposure to PA14, trained worms developed strong aversion to the pathogen, exhibiting a significantly positive choice index (**Figure 1** [↗](#)).

Critically, this avoidance behavior persisted in the untrained F1 and F2 progeny of trained worms, with both generations showing significantly higher choice indices compared to naïve controls (**Fig. 1** [↗](#), Table S1 and S2).

Our findings independently validate the transgenerational inheritance of learned pathogen avoidance initially reported by the Murphy lab. In contrast, Gainey et al. (2025) [↗](#) using a modified protocol failed to observe such inheritance beyond F1 despite multiple attempts. While we cannot definitively reconcile these differences, our results suggest that under tightly controlled

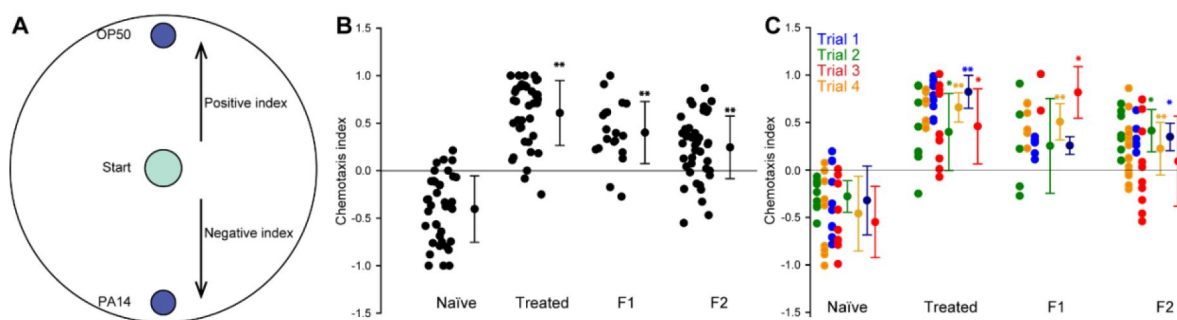


Figure 1.

Transgenerational inheritance of PA14 avoidance in *C. elegans*.

A. Chemotaxis assay setup. A 25 μ L drop of OP50 and PA14 bacteria of the same optical density ($OD_{600} = 1$) plus 1.0 μ L of 400 mM Na₃ were placed equidistant from the center of a 10cm chemotaxis plate where worms were placed at the start of the assay (see methods). indices were measured for naïve worms, trained worms, first-generation (F1), and second-generation (F2) offspring. **B.** Pooled results from four independent trials showing that Trained worms, which were exposed to PA14 for 24 hours, exhibited significantly higher choice indices, indicating learned avoidance of PA14. This aversion persisted in F1 and F2 progeny, demonstrating transgenerational inheritance of pathogen avoidance behavior. **C.** Comparison of data separated into individual trials. Kruskal-Wallis one-way ANOVA followed by Dunn's post hoc, * $p < 0.05$, ** $p < 0.01$ means and SEM shown (see Table S1 for statistics and Table S2 for raw data).

conditions, including bacterial lawn density, $OD_{600}=1.0$ standardization, and immobilization with sodium azide restricted to one hour (to capture initial preference behaviors), transgenerational inheritance through F2 is both detectable and statistically robust.

This underscores the assay's sensitivity to environmental variables, such as synchronization method and bacterial lawn density. This highlights the importance of consistency across experimental setups and support the view that context-dependent variation may underlie previously reported discrepancies.

These results confirm that, under tightly controlled conditions, transgenerational inheritance of PA14 avoidance extends through the F2 generation, though the response attenuates without further reinforcement.

Methods

Nematode Strains and Maintenance

Wild-type *Caenorhabditis elegans* (N2) were maintained on nematode growth medium (NGM) plates seeded with *Escherichia coli* OP50 at 20°C under standard conditions (Brenner 1974 [DOI](#)). Worm populations were passaged regularly to avoid overcrowding and starvation. OP50 was obtained from the *Caenorhabditis* Genetics Center (CGC).

Bacterial Cultures

We inoculated 6 mL of LB medium (pH 7.5) with single colonies of OP50 or *Pseudomonas aeruginosa* PA14 and incubated them overnight at 37°C with shaking at 250 rpm. For PA14, cultures did not exceed 16 hours of incubation or develop visible biofilms. For both OP50 and PA14, the overnight culture was diluted to an OD_{600} of 1.0 in fresh LB media prior to seeding.

Synchronization of Worms

Gravid adults were collected from OP50-seeded NGM plates by washing with liquid NGM buffer. We used liquid NGM buffer instead of M9 buffer (as specified in the original Murphy protocol) to maintain chemical consistency with the solid NGM culture plates. This modification minimizes potential osmotic stress since liquid NGM matches the pH (6.0) and ionic composition of the growth medium, whereas M9 buffer has a different pH (7.0) and ionic profile. After settling by gravity, a standard bleach solution was added and gently nutated for 5-10 minutes. The embryo pellet was washed 2-3 times with liquid NGM after centrifugation. Liquid NGM buffer is identical to agar NGM plates except that it does not include agar, peptone, cholesterol, or OP50 *E. coli*. The key difference between liquid NGM and M9 (aside from their pH) is that NGM has $CaCl_2$ and that M9 typically has more Na^+ while NGM has more K^+ due to the phosphate salts used. Approximately 250-350 eggs were plated onto each OP50-seeded NGM plate and incubated at 20°C for 48-52 hours until reaching late L4 stage.

Training and Choice Assay

Training plates (10 cm NGM) were seeded with 1 mL of overnight bacterial cultures and incubated at 25°C for 2 days. Choice assay plates (6 cm NGM) were prepared with 25 μ L spots of OP50 and PA14 on opposite sides placed 24 hrs prior to the assays. Late L4 worms were transferred to training plates (OP50 or PA14) for 24 hours at 20°C. For choice assays, 1.0 μ L of 400 mM sodium azide was placed on each bacterial spot before adding worms. After one hour, the number of worms immobilized on each spot was counted to calculate a Choice Index (CI):

$$CI = (\text{worms on OP50} - \text{worms on PA14}) / (\text{worms on OP50} + \text{worms on PA14})$$

We used 400 mM sodium azide rather than the 1 M concentration reported by [Moore et al. \(2019\)](#) [\[link\]](#) because preliminary trials showed that higher concentrations caused premature paralysis before worms could reach either bacterial spot, potentially biasing choice measurements. The 400 mM concentration provided sufficient immobilization while preserving the behavioral choice window.

Transgenerational Testing

Synchronized F1 progeny were obtained by bleaching trained adult worms and allowing embryos to develop on OP50-seeded plates at 20°C. On Day 1 of adulthood, F1 worms were tested using the same choice assay. The F2 generation was derived from untrained F1 adults, synchronized in the same manner, and tested in identical conditions.

Each behavioral assay was conducted using animals from a biologically independent growth plate. While F2 plates were derived from pooled embryos from multiple F1 parents, each assay represents an independent biological replicate with no reuse of animals across assays. F2 assays (n=45) exceeded F1 assays (n=20) due to PA14-induced fecundity reduction in trained worms, limiting the number of viable F1 progeny. The higher number of F2 assays reflects the greater reproductive success of healthy F1 animals and provides additional statistical power for population-level behavioral comparisons.

Controls and Additional Considerations

The data presented in this study are the result of four independent rounds of experimentations. Each individual assay was performed with animals harvested from unique culture plates, ensuring biological independence across all behavioral measurements. An average of 62 ± 43 animals participated in each assay (were immobilized in the NaN_3 and tallied at the end of the assay). We conducted an average of 8.5 assays per condition during each of the four replicates. Each replicate was a biologically independent experiment conducted on a different day. Our experimental design employed population-level comparisons across generations using unpaired statistical analyses, with no attempt to track individual lineages across generations. All plates used in the study were prepared at least two days before experiments to ensure consistent bacterial lawn growth. Worms were monitored to prevent starvation or overcrowding, as these conditions could influence their bacterial preferences and behavioral responses. Fresh bacterial stocks were maintained at -80°C and streaked weekly to ensure culture consistency.

Statistical Analysis

All statistical analyses were performed using SigmaPlot 14 (Inpixon). Outliers were identified and removed using the interquartile range criterion ($>1.5 \times \text{IQR}$), resulting in the exclusion of 5 out of 143 data points (3.5%). Their removal did not alter the significance or direction of reported effects. When datasets passed both the Shapiro-Wilk test for normality and the Brown-Forsythe test for equal variance, we used one-way ANOVA followed by Holm-Sidak post hoc comparisons versus the control group. For datasets that failed either assumption, we used Kruskal-Wallis one-way analysis of variance on ranks, followed by Dunn's post hoc comparisons versus control. The full statistical outputs are provided in Table S1, and raw data for individual assays are presented in Table S2.

Supplementary Information

TRIAL	Naïve	Trained	F1	F2	H	DF	P	Dunn's Naïve - Trained			Dunn's Naïve - F1			Dunn's Naïve - F2		
	N	N	N	N				Diff Rank	Q	P	Diff Rank	Q	P	Diff Rank	Q	P
1	10	12	6	7	29.23	3	<0.001	23.5	5.4	<0.001	9.7	1.8	0.2	12.3	2.4	0.044
2	8	8	5	7	12.8	3	0.005	12.4	3	0.008	10.1	2.2	0.09	13.1	3.1	0.006
3	10	10	2	11	18	3	<0.001	16.6	3.8	<0.001	21.4	2.9	0.013	10	2.4	0.053
					Shapiro-Wilk	Brown-Forsythe	Alpha-Power	Holm-Sidak Naïve - Trained			Holm-Sidak Naïve - F1			Holm-Sidak Naïve - F2		
								Diff Mean	t	P	Diff Mean	t	P	Diff Mean	t	P
4	10	8	6	18	0.48	0.084	0.05/1.00	1.1	8.3	<0.001	0.97	6.6	<0.001	0.68	6.1	<0.001
	Naïve	Trained	F1	F2	H	DF	P	Dunn's Naïve - Trained			Dunn's Naïve - F1			Dunn's Naïve - F2		
								Diff Rank	Q	P	Diff Rank	Q	P	Diff Rank	Q	P
Pooled	38	38	19	43	77.6	3	<0.001	78.4	8.5	<0.001	60.6	5.4	<0.001	46.9	5.3	<0.001

Table S1.

Summary of statistical analysis.

H: H-statistic; DF: degrees of freedom; Q: standardized difference between ran sums; Trial: Independent trials conducted on different dates; N: independent assays conducted on a trial (each trial derived from individual culture plate); **Naïve**: Day-1 adults without previous exposure to PA14; **Trained**: Day-1 adults with previous exposure to PA14; **F1**: Day-1 adults without previous exposure to PA14 derived from **Trained** group; **F2**: Day-1 adults without previous exposure to PA14 derived from **F1** group.

Generation	Trial	Assay	PA14_count	OP50_count	Choice_Index
Naïve	1	1	19	24	0.12
Naïve	1	2	115	49	-0.4
Naïve	1	3	24	37	0.21
Naïve	1	4	45	56	0.11
Naïve	1	5	24	12	-0.33
Naïve	1	6	84	23	-0.57
Naïve	1	7	41	33	-0.11
Naïve	1	8	34	9	-0.58
Naïve	1	9	37	5	-0.76
Naïve	1	10	97	18	-0.69
Naïve	2	1	32	14	-0.39

Table S2.

Raw data used in this study.

Naïve	2	2	73	54	-0.15
Naïve	2	3	71	44	-0.23
Naïve	2	4	7	30	0.62
Naïve	2	5	16	88	0.69
Naïve	2	6	95	48	-0.33
Naïve	2	7	47	37	-0.12
Naïve	2	8	84	38	-0.38
Naïve	2	9	79	22	-0.56
Naïve	2	10	23	20	-0.07
Naïve	3	1	5	2	-0.43
Naïve	3	2	26	23	-0.06
Naïve	3	3	15	11	-0.15
Naïve	3	4	48	7	-0.75
Naïve	3	5	32	7	-0.64
Naïve	3	6	18	2	-0.8
Naïve	3	7	16	16	0
Naïve	3	8	18	0	-1
Naïve	3	9	7	1	-0.75
Naïve	3	10	7	0	-1
Naïve	4	1	15	8	-0.3
Naïve	4	2	181	21	-0.79
Naïve	4	3	48	3	-0.88
Naïve	4	4	22	2	-0.83
Naïve	4	5	28	13	-0.37
Naïve	4	6	13	7	-0.3
Naïve	4	7	22	22	0
Naïve	4	8	43	0	-1
Naïve	4	9	30	24	-0.11
Naïve	4	10	22	26	0.08
Trained	1	1	3	16	0.68
Trained	1	2	1	48	0.96
Trained	1	3	1	7	0.75
Trained	1	4	2	18	0.8
Trained	1	5	7	24	0.55
Trained	1	6	2	42	0.91
Trained	1	7	0	29	1
Trained	1	8	41	20	-0.34
Trained	1	9	0	71	1
Trained	1	10	19	62	0.53
Trained	1	11	0	97	1

Table S2. (continued)

Trained	1	12	2	81	0.95
Trained	1	13	7	126	0.89
Trained	2	1	2	34	0.89
Trained	2	2	8	47	0.71
Trained	2	3	25	4	-0.72
Trained	2	4	4	65	0.88
Trained	2	5	29	43	0.19
Trained	2	6	6	16	0.45
Trained	2	7	50	30	-0.25
Trained	2	8	56	74	0.14
Trained	2	9	9	13	0.18
Trained	3	1	8	15	0.3
Trained	3	2	21	181	0.79
Trained	3	3	3	48	0.88
Trained	3	4	2	22	0.83
Trained	3	5	13	28	0.37
Trained	3	6	7	13	0.3
Trained	3	7	22	22	0
Trained	3	8	0	43	1
Trained	3	9	24	30	0.11
Trained	3	10	26	22	-0.08
Trained	4	1	3	32	0.83
Trained	4	2	8	47	0.71
Trained	4	3	5	32	0.73
Trained	4	4	10	30	0.5
Trained	4	5	13	34	0.45
Trained	4	6	6	40	0.74
Trained	4	7	1	13	0.86
Trained	4	8	4	13	0.53
F1	1	1	16	30	0.3
F1	1	2	21	31	0.2
F1	1	3	19	41	0.4
F1	1	4	24	17	-0.2
F1	1	5	17	22	0.1
F1	1	6	12	23	0.3
F1	1	7	21	42	0.3
F1	2	1	2	42	0.9
F1	2	2	7	11	0.2
F1	2	3	5	19	0.6
F1	2	4	27	19	-0.2

Table S2. (continued)

F1	2	5	28	16	-0.3
F1	3	1	0	29	1
F1	3	2	5	21	0.6
F1	4	1	10	60	0.7
F1	4	2	11	45	0.6
F1	4	3	19	44	0.4
F1	4	4	15	38	0.4
F1	4	5	5	29	0.7
F1	4	6	13	21	0.2
F2	1	1	0	57	1
F2	1	2	37	76	0.35
F2	1	3	30	53	0.28
F2	1	4	17	31	0.29
F2	1	5	10	46	0.64
F2	1	6	10	15	0.2
F2	1	7	18	37	0.35
F2	1	8	8	21	0.45
F2	1	9	50	39	-0.12
F2	2	1	8	29	0.57
F2	2	2	20	43	0.37
F2	2	3	11	17	0.21
F2	2	4	8	34	0.62
F2	2	5	9	18	0.33
F2	2	6	8	45	0.7
F2	2	7	18	22	0.1
F2	3	1	4	26	0.73
F2	3	2	18	12	-0.2
F2	3	3	68	79	0.07
F2	3	4	14	90	0.73
F2	3	5	38	11	-0.55
F2	3	6	58	21	-0.47
F2	3	7	29	67	0.4
F2	3	8	23	24	0.02
F2	3	9	54	41	-0.14
F2	3	10	73	37	-0.33
F2	3	11	7	31	0.63
F2	4	1	30	62	0.35
F2	4	2	36	67	0.3
F2	4	3	18	257	0.87
F2	4	4	20	36	0.29

Table S2. (continued)

Table S2. (continued)

F2	4	5	26	45	0.27
F2	4	6	16	41	0.44
F2	4	7	6	31	0.68
F2	4	8	22	24	0.04
F2	4	9	24	31	0.13
F2	4	10	58	56	-0.02
F2	4	11	71	48	-0.19
F2	4	12	24	51	0.36
F2	4	13	57	75	0.14
F2	4	14	62	62	0
F2	4	15	35	74	0.36
F2	4	16	94	69	-0.15
F2	4	17	75	68	-0.05
F2	4	18	20	46	0.39

Table S3.

Reagents and suppliers used in this study.

Reagent	Supplier	Catalog
NGM Agar ingredients		
Agar	Carolina	842133
Cholesterol	FischerSci	AAA1147018
MgSO ₄	FischerSci	18-604-121
CaCl ₂	FischerSci	18-601-250
KH ₂ PO ₄	FischerSci	01-337-803
K ₂ HPO ₄	FischerSci	01-337-798
NaCl	FischerSci	BP358-212
Peptone	FischerSci	BP1420-500
Liquid NGM Buffer		
MgSO ₄	FischerSci	18-604-121
CaCl ₂	FischerSci	18-601-250
KH ₂ PO ₄	FischerSci	01-337-803
K ₂ HPO ₄	FischerSci	01-337-798
NaCl	FischerSci	BP358-212
LB Broth		
Yeast Extract	FischerSci	BP1422-500
Tryptone	FischerSci	BP1421-500
H ₂ O (purified)	Barnstead Genepure	10-451-061
Paralytic		
NaN ₃	FischerSci	S0489100G
Bleach solution		
NaOH	FischerSci	S318-100
NaOCl (6%)	Total Home Bleach	N/A

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Additional information

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Reviewer #1 (Public review):

Summary:

The manuscript addresses the discordant reports of the Murphy (Moore et al., 2019; Kaletsky et al., 2020; Sengupta et al., 2024) and Hunter (Gainey et al., 2025) groups on the existence (or robustness) of transgenerational epigenetic inheritance (TEI) controlling learned avoidance of *C. elegans* to *Pseudomonas aeruginosa*. Several papers from Colleen Murphy's group describe and characterize *C. elegans* transgenerational inheritance of avoidance behaviour. In the hands of the Murphy group, the learned avoidance is maintained for up to four generations, however, Gainey et al. (2025) reported an inability to observe inheritance of

learned avoidance beyond the F1 generation. Of note, Gainey et al used a modified assay to measure avoidance, rather than the standard assay used by the Murphy lab. A response from the Murphy group suggested that procedural differences explained the inability of Gainey et al.(2025) to observe TEI. They found two sources of variability that could explain the discrepancy between studies: the modified avoidance assay and bacterial growth conditions (Kaletsky et al., 2025). The standard avoidance assay uses azide as a paralytic to capture worms in their initial decision, while the assay used by the Hunter group does not capture the worm's initial decision but rather uses cold to capture the location of the population at one point in time.

In this short report, Akinosho, Alexander, and colleagues provide independent validation of transgenerational epigenetic inheritance (TEI) of learned avoidance to *P. aeruginosa* as described by the Murphy group by demonstrating learned avoidance in the F2 generation. These experiments used the protocol described by the Murphy group, demonstrating reproducibility and robustness.

Strengths:

Despite the extensive analyses carried out by the Murphy lab, doubt may remain for those who have not read the publications or for those who are unfamiliar with the data, which is why this report from the Vidal-Gadea group is so important. The observation that learned avoidance was maintained in the F2 generation provides independent confirmation of transgenerational inheritance that is consistent with reports from the Murphy group. It is of note that Akinosho, Alexander et al. used the standard avoidance assay that incorporates azide, and followed the protocol described by the Murphy lab, demonstrating that the data from the Moore and Kaletsky publications are reproducible, in contrast to what has been asserted by the Hunter group.

Comments on revised version:

I am happy with the responses to reviews.

<https://doi.org/10.7554/eLife.107034.2.sa3>

Reviewer #2 (Public review):

Summary:

The manuscript "Independent validation of transgenerational inheritance of learned pathogen avoidance in *C. elegans*" by Akinosho and Vidal-Gadea offers evidence that learned avoidance of the pathogen PA14 can be inherited for at least two generations. In spite of initial preference for the pathogen when exposed in a 'training session', 24 hours of feeding on this pathogen evoked avoidance. The data are robust, replicated in 4 trials, and the authors note that diminished avoidance is inherited in generations F1 and F2.

Strengths:

These results contrast with those reported by Gainey et al, who only observed intergenerational inheritance for a single generation. Although the authors' study does not explain why Gainey et al fail to reproduce the Murphy lab results, one possibility is that a difference in a media ingredient could be responsible.

Comments on revised version:

The responses to the reviewer comments appear reasonable for the most part.

<https://doi.org/10.7554/eLife.107034.2.sa2>

Reviewer #3 (Public review):

Summary:

This short paper aims to provide an independent validation of the transgenerational inheritance of learned behaviour (avoidance) that has been published by the Murphy lab. The robustness of the phenotype has been questioned by the Hunter lab. In this paper, the authors present one figure showing that transgenerational inheritance can be replicated in their hands. Overall, it helps to shed some light on a controversial topic.

Strengths:

The authors clearly outline their methods, particularly regarding the choice of assay, so that attempting to reproduce the results should be straightforward. It is nice to see these results repeated in an independent laboratory.

Comments on revised version:

I'm happy with the response to reviewers.

<https://doi.org/10.7554/eLife.107034.2.sa1>

Author response

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):*Confirmation of daf-7::GFP data and inheritance beyond F2*

Reviewer suggested confirming daf-7::GFP molecular marker data and testing inheritance beyond the F2 generation to further strengthen the findings.

We agree these experiments would provide valuable mechanistic insights into the molecular basis of transgenerational inheritance. However, our study was specifically designed as a reproducibility study focusing on the central controversy regarding F2 inheritance (Gainey et al. vs. Murphy lab findings). The daf-7::GFP molecular marker experiments, while important for understanding mechanisms, represent a different research question requiring extensive additional resources and expertise beyond the scope of this validation study. Our primary goal was to provide independent confirmation of the disputed F2 inheritance using standardized behavioral assays. It is our hope that future work will pursue these important mechanistic validations.

"Exhaustive attempts" language

Reviewer disagreed with characterizing Gainey et al.'s efforts as "exhaustive attempts" since they modified the original protocol.

We revised this statement in the Results and Discussion to more accurately reflect the experimental situation: "In contrast, Gainey et al. (2025), representing the Hunter group, reported that while parental and F1 avoidance behaviors were evident, transgenerational inheritance was not reliably observed beyond the F1 generation under their experimental conditions."

Importance of sodium azide

Reviewer suggested including more discussion about the recent findings on the importance of sodium azide in the assay, referencing the Murphy group's response paper.

We have prominently highlighted the critical role of sodium azide in our Introduction with strengthened language that emphasizes its importance for resolving the scientific controversy: "Critically, Kaletsky et al. (2025) demonstrated that omission of sodium azide during scoring can completely abolish detection of inherited avoidance, revealing that this key methodological difference may explain the conflicting results between laboratories. The use of sodium azide to immobilize worms at the moment of initial bacterial choice appears essential for capturing the inherited behavioral response. These findings highlight how seemingly minor methodological variations can dramatically impact detection of transgenerational inheritance and underscore the need for independent replication using standardized protocols."

| *Protocol fidelity statement*

Reviewer requested a more direct statement clarifying that we followed the Murphy group protocol, noting that we made some modifications.

We followed the core Murphy lab protocol with two evidence-based optimizations that preserve the essential experimental elements: 1) We used 400 mM sodium azide instead of 1 M based on preliminary data showing the higher concentration caused premature paralysis before worms could make behavioral choices, and 2) We used liquid NGM buffer instead of M9 to maintain chemical consistency with the solid NGM plates used for worm culture, minimizing potential osmotic stress. These modifications improved experimental reliability while maintaining the critical components: sodium azide immobilization, bacterial lawn density standardization ($OD_{600} = 1.0$), and synchronized scoring conditions that are essential for detecting inherited avoidance.

| *Overstated dilution claim*

Reviewer noted that the statement about "gradual decrease" in avoidance strength was overstated and didn't reflect the actual data presented in the manuscript.

We removed this statement.

| *Environmental variables phrasing*

Reviewer found the sentence about environmental variables unclear, noting that Gainey et al. didn't actually acknowledge variability but saw it as indicating error or stochastic processes.

We refined this statement for greater precision and clarity: "This underscores the assay's sensitivity to environmental variables, such as synchronization method and bacterial lawn density. This highlights the importance of consistency across experimental setups and support the view that context-dependent variation may underlie previously reported discrepancies."

| **Reviewer #2 (Public Review):**

| *Reagent sourcing*

Reviewer suggested listing the sources of media ingredients with company names and catalog numbers, as this might be important for reproducibility.

To ensure complete reproducibility, we created a comprehensive Table S3 listing all reagents, suppliers, and catalog numbers used in our experiments. This detailed information enables exact replication of our experimental conditions and addresses potential variability that might arise from different reagent sources between laboratories.

Reviewer #3 (Public Review):

Raw data transparency

Reviewer noted that while a spreadsheet with choice assay results was provided, the individual raw data from assays was not included, which would be helpful for assessing sample sizes.

We now provide complete experimental transparency through Table S2, which contains individual choice indices from all 138 assays conducted across four independent trials. This comprehensive dataset allows full assessment of our experimental outcomes, statistical robustness, and reproducibility while enabling other researchers to perform independent statistical analyses.

F1/F2 assay disparity

Reviewer questioned whether the higher number of F2 assays compared to F1 represented truly independent assays, asking if multiple F2 assays were performed from offspring of one F1 plate (which would not represent independent assays).

We clarified this important statistical consideration in Methods (Transgenerational Testing): "Each behavioral assay was conducted using animals from a biologically independent growth plate. While F2 plates were derived from pooled embryos from multiple F1 parents, each assay represents an independent biological replicate with no reuse of animals across assays. F2 assays (n=45) exceeded F1 assays (n=20) due to PA14-induced fecundity reduction in trained worms, limiting the number of viable F1 progeny. The higher number of F2 assays reflects the greater reproductive success of healthy F1 animals and provides additional statistical power for population-level behavioral comparisons." We also enhanced our Controls section to clarify that "Our experimental design employed population-level comparisons across generations using unpaired statistical analyses, with no attempt to track individual lineages across generations."

Methodological variations overstatement

Reviewer felt the Introduction overstated the findings by suggesting the authors "address potential methodological variations," when they only used one assay setup throughout.

We have corrected the Introduction to accurately reflect our study design and scope: "Here, we adapted the protocol established by the Murphy group, maintaining the critical use of sodium azide to paralyze worms at the time of choice, to test whether parental exposure to PA14 elicits consistent avoidance in subsequent generations. Our study specifically focuses on the transmission of learned avoidance through the F2 generation, beyond the intergenerational (F1) effect, because this is where divergence between published studies begins."

Reviewer #1 (Recommendations for the authors):

Worm numbers

Reviewer noted that information about the number of worms used should be included in the training and choice assay methods section rather than separated.

We clarified worm numbers and sample sizes in the Methods (Controls and Additional Considerations): "Each individual assay averaged 62 ± 43 animals (range: 15-150 worms per assay), with a total of 138 assays conducted across four independent experimental trials. The variation in worm numbers per assay reflects natural variation in worm recovery and immobilization efficiency during choice assays. We conducted an average of 8.5 assays per condition during each of the four replicates."

| *Figure 1 legend and consistency*

Reviewer identified several issues: inconsistent terminology ("treated" vs "trained"), incorrect statistical test naming, missing p-value annotations, and need for consistency between figure and legend. We have systematically addressed all figure consistency and statistical annotation issues:

| *Replaced inconsistent "treated" terminology with "trained" throughout*

Corrected the statistical test description to accurately reflect our analysis: "Kruskal-Wallis oneway ANOVA followed by Dunn's post hoc" which properly corresponds to the statistical tests detailed in Table S1

Added explicit p-value annotations in the figure legend: " $*p < 0.05$, $**p < 0.01$ means and SEM shown (see Table S1 for statistics and Table S2 for raw data)"

Ensured consistent terminology between figure and legend

| *NGM vs. M9 buffer*

Reviewer questioned whether we used NGM buffer or M9 buffer for washing steps, noting that NGM isn't usually referred to as "buffer."

We have prominently featured and thoroughly clarified our rationale for using liquid NGM buffer in the Methods (Synchronization of Worms section). The explanation now appears upfront in the methods: "We used liquid NGM buffer instead of M9 buffer (as specified in the original Murphy protocol) to maintain chemical consistency with the solid NGM culture plates. This modification minimizes potential osmotic stress since liquid NGM matches the pH (6.0) and ionic composition of the growth medium, whereas M9 buffer has a different pH (7.0) and ionic profile." We provide detailed chemical differences and explain that this modification maintains consistency with culture conditions while preserving essential experimental procedures.

| *Grammar/typos*

Reviewer noted that the manuscript needed thorough proofreading to address grammatical errors and typographical mistakes.

We have conducted comprehensive proofreading and editing throughout the manuscript to resolve grammatical and typographical errors. Specific improvements include: clarified sentence structure in the Introduction and Results sections, corrected technical terminology consistency, improved figure legend clarity, and enhanced overall readability while maintaining scientific precision.

| *Sodium azide concentration*

Reviewer noted that our sodium azide concentration differed from the Moore paper and requested comment on this difference.

We have included explicit justification for our sodium azide concentration choice in the Methods (Training and Choice Assay): "We used 400 mM sodium azide rather than the 1 M concentration reported by Moore et al. (2019) because preliminary trials showed that higher concentrations caused premature paralysis before worms could reach either bacterial spot, potentially biasing choice measurements. The 400 mM concentration provided sufficient immobilization while preserving the behavioral choice window."

Reviewer #2 (Recommendations for the authors):

Comparative reagent analysis

Reviewer suggested creating a supplemental table comparing reagent sources between our study, Gainey et al., and Murphy et al., proposing that media ingredient differences might explain the discrepancies.

While direct reagent comparison between laboratories was beyond the scope of this validation study, we recognize this as an important consideration for understanding experimental variability. Our comprehensive reagent sourcing information (Table S3) provides the foundation for future comparative studies. We encourage collaborative efforts to systematically compare reagent sources across laboratories, as media component differences could contribute to the experimental variability observed between research groups. Such analyses would be valuable for establishing standardized protocols across the field.

Conclusion

We hope that these revisions satisfactorily address the reviewers' concerns. We believe these improvements significantly strengthened the manuscript's contribution to resolving this important scientific controversy.

We thank the reviewers again for their invaluable insights and constructive feedback, which have substantially improved the quality and impact of our work.

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